The binomial theorem is true for all real numbers n. For example, when $n = \sqrt{2}$, we have

$$(1+x)^{\sqrt{2}} = 1 + \sqrt{2}x + \frac{\sqrt{2}(\sqrt{2}-1)}{2!}x^2 + \frac{\sqrt{2}(\sqrt{2}-1)(\sqrt{2}-2)}{3!}x^3 + \cdots, \quad |x| < 1.$$

5.6.6 Exponential Series

The series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ is called an ***exponential series***. It can be proved that this series converges for all values of x.

For any real number x, $\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$ where

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots.$$

We have

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots \tag{5.1}$$

for all values of x. By taking $-x$ in place of x in (5.1) we get

$$e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \cdots. \tag{5.2}$$

In particular,

$$\frac{1}{e} = e^{-1} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots.$$

From (5.1) and (5.2) we get

$$\frac{e^x + e^{-x}}{2} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots \text{ and } \frac{e^x - e^{-x}}{2} = \frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$$

In particular we have

$$\frac{e + e^{-1}}{2} = 1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \cdots \text{ and } \frac{e - e^{-1}}{2} = \frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \cdots$$

By taking $2x$ in place of x in (5.1) we get

$$e^{2x} = 1 + \frac{2x}{1!} + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \frac{(2x)^4}{4!} + \cdots.$$

On simplification we see that

$$e^{2x} = 1 + \frac{2x}{1!} + \frac{4x^2}{2!} + \frac{8x^3}{3!} + \frac{16x^4}{4!} + \cdots.$$